

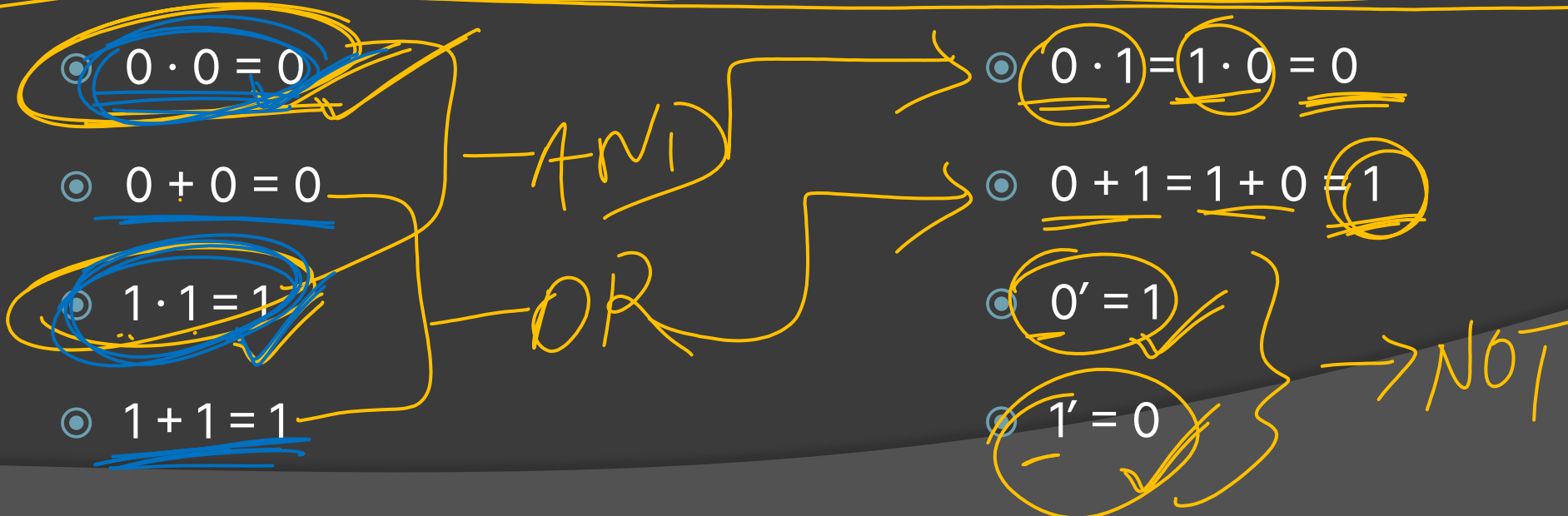
BOOLEAN FUNCTIONS & BOOLEAN LAWS - 2

Unit 2 ✓ Digital Electronics

Boolean Algebra Axioms (postulates)

An axiom is nothing more than the definition of three basic logic operations (AND, OR and NOT).

All axioms defined in Boolean algebra are the results of an operation that is performed by a logical gate.



Laws of Boolean Algebra based on Axioms



Commutative Laws:

✓ $A + B = B + A$, and $A \cdot B = B \cdot A$

OR

AND

Identity Laws:

✓ $A + 0 = A$

✓ $A \cdot 1 = A$

Annulment Laws:

✓ $A + 1 = 1$

✓ $A \cdot 0 = 0$

⇒ NULL Laws

$\Rightarrow b c \Rightarrow A \Rightarrow \underline{1} \text{ or } \underline{0}$

$1 + 1 = 1$

$0 + 1 = 1$

■ Complement Laws:

✓ If $A + A' = 1$

✓ If $A \cdot A' = 0$

✓ **Double Complement (Involution):** $(A')' = A$

■ Associative Laws:

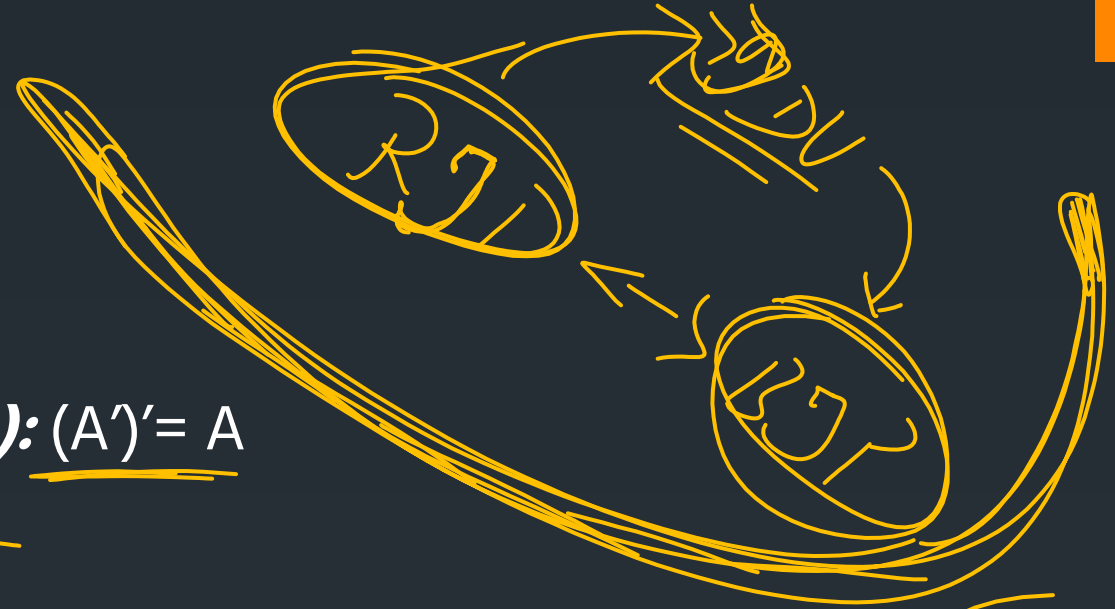
✓ $(A + B) + C = A + (B + C)$

✓ $(A \cdot B) \cdot C = A \cdot (B \cdot C)$

AND

OR

$Bihar \Rightarrow NDA \Rightarrow BIP + JDU$
 $A \cdot G \cdot B = JDU + RJD$



◉ Distributive Laws:

✓ $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

✓ $A + B \cdot C = (A + B) \cdot (A + C)$

◉ Idempotent Law:

✓ $A \cdot A = A$, If $A=1$, then $A \cdot A = 1 \cdot 1 = 1 = A$ and if $A=0$, then $A \cdot A = \underline{0 \cdot 0} = \underline{0} = A$

✓ $A + A = A$, If $A=1$, then $A + A = 1 + 1 = 1 = A$ and if $A=0$, then $A + A = 0 + 0 = 0 = A$

◉ Absorption Law:

✓ $A + A \cdot B = A$

✓ $A \cdot (A + B) = A$

$\Rightarrow A \cdot (1 + B) \Rightarrow A \cdot 1 \Rightarrow A$

$\Rightarrow (A \cdot A) + (A \cdot B) \Rightarrow (A + A \cdot B) = A$



De-Morgan's Laws:

✓ $(A + B)' = A' \cdot B'$

✓ $(A \cdot B)' = A' + B'$

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$$(A + B)' \Rightarrow A' \oplus B'$$
$$\Rightarrow A' \cdot B'$$

$$(A \cdot B)' \Rightarrow A' \cdot B'$$
$$= A' + B'$$

TRUTH TABLE FOR IDENTITY LAWS

A	$A + 0$
0	0
1	1

A	$A \cdot 1$
0	0
1	1

Conclusion:

From the truth tables, it is verified that:

$A + 0 = A$ and $A \cdot 1 = A$. Hence, the **Identity Laws of Boolean Algebra (Over OR & AND)** are proved.

Truth Table for Annulment Laws



A	$A + 1$
✓ 0	1
✓ 1	1

Handwritten annotations: Blue circles around the '1' values in the $A + 1$ column. Arrows point from the circles to the expressions $0 + 1$ and $1 + 1$ written to the right of the table.

A	$A \cdot 0$
<u>0</u>	0
1	0

Handwritten annotations: Blue circles around the '0' values in the $A \cdot 0$ column. Arrows point from the circles to the expressions $0 \cdot 0$ and $1 \cdot 0$ written to the right of the table.

Conclusion:

From the truth tables, it is verified that:

$A + 1 = 1$ and $A \cdot 0 = 0$. Hence, the **Annulment Laws of Boolean Algebra (Over OR & AND)** are proved.

Truth Table for Idempotent Laws

OR

<u>A</u>	<u>A + A</u>
<u>0</u>	0 $\leftarrow 0 + 0$
<u>1</u>	1 $\leftarrow 1 + 1$

AND

<u>A</u>	<u>A · A</u>
0	0 $\leftarrow 0 \cdot 0$
1	1 $\leftarrow 1 \cdot 1$

Conclusion:

- From the truth tables, it is verified that:
- $A + A = A$ and $A \cdot A = A$. Hence, the **Idempotent Laws of Boolean Algebra (Over OR & AND)** are proved.

TRUTH TABLE FOR COMPLEMENT LAWS

A	A'	A + A'
0	1	1
1	0	1

A	A'	A · A'
0	1	0
1	0	0

Conclusion:

From the truth tables, it is verified that:

$A + A' = 1$ and $A \cdot A' = 0$. Hence, the **Idempotent Laws of Boolean Algebra (Over OR & AND)** are proved.

Proof of De Morgan's First Theorem

	<u>A</u>	<u>B</u>	<u>A'</u>	<u>B'</u>	<u>A · B</u>	<u>(A · B)'</u>	<u>A' + B'</u>
1	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	1	<u>1</u>
2	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	1	<u>1</u>
3	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	1	<u>1</u>
4	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	0	<u>0</u>

$\hookrightarrow 2 \text{ I/O} \Rightarrow 2^2 = 4 \text{ comb's}$

Conclusion:

- From the last two columns, it is verified that:
- $(A \cdot B)' = A' + B'$ Hence, **De Morgan's First Theorem** is proved.

$\Rightarrow (A + B)' = \underline{A' \cdot B'} \Leftarrow \underline{H/W}$

Truth Table for Absorption Law

A	B	AB	$A + A \cdot B$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Conclusion :

- From the truth tables $A \cdot (A + B) = A$, **First Absorption Law of Boolean Algebra** are verified.

$$2^{nd} \Rightarrow A \cdot (A + B) = A \Rightarrow \text{H/W}$$

$$H/S \Rightarrow A + (B \cdot C) = (A + B) \cdot (A + C)$$

Truth Table for First Distributive Law

A	B	C	B + C	A(B + C)	AB	AC	AB + AC
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

Conclusion:

$\Rightarrow 3 \text{ IP} \Rightarrow 2^3 = 8 \text{ (combs)}$

Compare

- From the last two columns, it is observed that:
- $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$ Hence, the **Distributive Law of Boolean Algebra** is verified.

Truth Table for Associative Law

A	B	C	A + B	<u>(A + B) + C</u>	B + C	<u>A + (B + C)</u>
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	1	0	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Conclusion:

- From the last two columns, it is clear that:
- $(A + B) + C = A + (B + C)$ Hence, the **Associative Law for OR operation** is verified.

$$H/W \Rightarrow (A \cdot B) \cdot C = A \cdot (B \cdot C)$$

Two-Variable Minterms

- Let the variables be A and B be the inputs. Total minterms = $2^2 = 4$.

$$A=0 \Rightarrow A', B=0 \Rightarrow B'$$

Input A	Input B	A · B (AND)	Minterm Index	Minterm Expression
0	0	0	m_0	$A' \cdot B'$
0	1	0	m_1	$A' \cdot B$
1	0	0	m_2	$A \cdot B'$
1	1	1	m_3	$A \cdot B$

minterm

Expression

2th Sci
+ BE/BTech

~~DD = TRB~~

~~UTU = 864/7~~

Canonical Sum of Minterms (SOP Form):

- Any Boolean expression can be written as a **sum (OR) of minterms** for which the function output is 1. *(Bcoz, Neutral value in Multiplication is 1).*
- Let, O/P of Some Boolean expression = 1 for inputs for minterms: m_1 and m_2 :
 - So, $F(A, B) = m_1 + m_2 = A' \cdot B + A \cdot B'$
- In canonical form, $F(A, B) = \sum m(1, 3) = A' \cdot B + A \cdot B'$

2 variable

$\Rightarrow$ for 2 var ITP. write canonical form of m_0, m_2, m_3
A, B -
X, Y -
with expression (SOP)

3 VARIABLE MINTERMS

Let the variables be A, B and C be 3 inputs. Total minterms = $2^3 = 8$.

A	B	C	Minterm <i>expression</i>	<i>minterm</i> Decimal Index
<u>0</u>	<u>0</u>	<u>0</u>	$A' \cdot B' \cdot C'$	m_0
0	0	1	$A' \cdot B' \cdot C$	m_1
0	1	0	$A' \cdot B \cdot C'$	m_2
0	1	1	$A' \cdot B \cdot C$	m_3
1	0	0	$A \cdot B' \cdot C'$	m_4
1	0	1	$A \cdot B' \cdot C$	m_5
1	1	0	$A \cdot B \cdot C'$	m_6
1	1	1	$A \cdot B \cdot C$	m_7

1
2
3
4
5
6
7
8

CANONICAL SUM OF MINTERMS (SOP FORM):

❖ Let the function **$F(A, B, C)$** be 1 for minterms: m_1, m_3, m_5, m_7

❖ So, $F(A, B, C) = m_1 + m_3 + m_5 + m_7 \leftarrow \text{SOP}$

❖ Canonical SOP Form:

$$F(A, B, C) = \sum m(1, 3, 5, 7) = A' \cdot B' \cdot C + A' \cdot B \cdot C + A \cdot B' \cdot C + A \cdot B \cdot C$$

Write canonical SOP form with expression for 3 variable

m_1, m_2, m_4, m_5 and m_6

A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

$m_0 \Rightarrow A'B'C'D'$
 $m_1 \Rightarrow A'B'C'D$

2-VARIABLE MAXTERMS

A	B	Maxterm Expression	Maxterm Index
0	0	$(A + B)$	M_0
0	1	$(A + B')$	M_1
1	0	$(A' + B)$	M_2
1	1	$(A' + B')$	M_3

Write Canonical SOP form for 4 variable min terms:
 $2^4 = 16 \Rightarrow m_0 \Rightarrow m_{15}$
 $m_2, m_7, m_9, m_{11}, m_{13}$